Max Duty, Minimum Overhead:

$\mathcal{O}(1)$-Memory Activation Probing and the Scaling Limits of Long-Context Safety Monitors

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Abstract

Activation probes are now deployed safety infrastructure, shipped inside frontier assistants and used as white-box members of untrusted-monitor ensembles. They are also memory-bound: an attention-pooled probe materialises a score tensor over the whole sequence, so its footprint grows with the one quantity an adversary controls, the context length. We study aggregation as the independent variable, holding the token transform fixed, and characterise a hard-maximum probe (*MultiMax*) against softmax attention pooling, mean pooling and full self-attention across $N \in [128, 131{,}072]$. Three results. **(i) Systems.** MultiMax streams its reduction and carries only per-head state, giving activation overhead of $\Theta(\min(C, N))$ in the chunk width C: measured at a flat 19.1 MiB from N=8,192 to 131,072 (α=0.000) against 652.1 MiB for softmax pooling (α=0.842), while self-attention reaches 10.1 GiB and exhausts memory. A chunk ablation confirms the constant is chosen, not free. **(ii) Optimisation.** The hard-max subgradient is supported on H tokens per sequence, and at weak signal this prevents learning entirely (0.00 recall at its own training length). Annealing a *normalised* Boltzmann operator $\tau \log(\frac{1}{N} \sum_j e^{s_j/\tau})$ to $\tau \to 0$ restores 1.00 recall; the normalisation is load-bearing, since plain LogSumExp injects a length-dependent $H\tau \log N$ offset. Calibration by Platt scaling reduces Brier from 0.0538 to 0.0122 and makes uncertainty gating usable at all. **(iii) Limits.** Against an adversary who fragments a fixed signal budget across m non-contiguous tokens, MultiMax and softmax pooling fail together at m=64; past that boundary softmax retains substantially better ranking (AUROC 0.716 vs 0.523 at m=256). We therefore do not claim a general detection advantage. The defensible claim is architectural: MultiMax is the right $\Theta(1)$-memory *first stage* of a cascade, not a replacement for inspection.

1 Introduction

Monitoring a frontier model's activations for misuse is no longer a research convenience. Probes are deployed in user-facing systems [14] and appear as white-box members inside AI-control protocols, where a cheap monitor gates an untrusted policy [8, 10]. Two properties then matter simultaneously, and they are usually studied apart: the probe must *detect*, and it must do so within a memory budget that does not grow with the context an adversary supplies.

That second requirement is sharp. Context windows now reach 10^5–10^6 tokens, and a guardrail that allocates memory proportional to the input is a guardrail whose cost the attacker sets. Kramár et al. [14] report that production probes fail under short-to-long distribution shift and that architecture choice, not merely training data, governs the outcome. Chen et al. [7] trace long-context guardrail failure to attention dilution. Neither isolates aggregation as a controlled variable.

We do. Fixing a shared token transform ϕ, we vary only the sequence reduction ρ, and compare four aggregators over five orders of magnitude in N. Our contributions:

- **A streaming hard-max probe with a measured memory law.** MultiMax carries (B, H) state across chunks, giving $\Theta(\min(C, N))$ activation overhead. We verify this with a chunk$\times$length ablation rather than asserting $\mathcal{O}(1)$ (§5.1, Rem. 5.1).
- **An optimisation obstruction and its fix.** We identify the subgradient support of the maximum as the reason hard-max probes can fail to train at weak signal, and resolve it with normalised Boltzmann annealing (§3).

- **An honest boundary.** We construct the adversary that specifically attacks a maximum, locate the failure at m=64, and report that softmax pooling degrades more gracefully past it. This bounds our own claim (§5.3).
- **A calibrated cascade.** Sum-of-maxima logits are upward-biased; we show temperature scaling is structurally insufficient and that budget-driven thresholding is the correct interface (§4).

All numbers below are read programmatically from a single benchmark artifact; a claims checker verifies every figure quoted in this paper against it.

2 Aggregation as the Independent Variable

Let $x_{i,j} \in \mathbb{R}^d$ be the residual-stream activation at layer ℓ and position $j \in \{1, \ldots, N\}$. A probe is

$$f_\theta(X_i) \;=\; \rho\big(\phi(x_{i,1}), \ldots, \phi(x_{i,N})\big), \qquad y_{i,j} = \phi(x_{i,j}) = \sigma(W_1 x_{i,j} + b_1) \in \mathbb{R}^m. \tag{1}$$

All probes share ϕ; only ρ differs. Writing $\alpha_j \geq 0$, $\sum_j \alpha_j = 1$ for a convex pooling weight, mean pooling sets $\alpha_j = 1/N$ and softmax attention sets $\alpha_j \propto \exp(q^\top y_j/\sqrt{m})$, both giving logit $w^\top(\sum_j \alpha_j y_j) + b$. MultiMax is not of this form:

$$\boxed{a_h(X_i) = \max_{1 \leq j \leq N} v_h^\top y_{i,j}, \qquad \text{logit}_i = \sum_{h=1}^{H} a_h(X_i) + b,} \tag{2}$$

with predictions through a bounded link $\hat{p} = \varsigma(\Pi_{[-c,c]}(\text{logit}))$, $\varsigma(z) = (1 + e^{-z})^{-1}$, $c = 10$.

2.1 The bounded link and gradient propagation under saturation

The clamp exists for a numerical reason. A MultiMax logit is a sum of H maxima, and by Remark 2.6 each summand grows like $\tau\sqrt{2\log N}$ even under the null, so $|z|$ drifts upward with context length. In fp16 the sigmoid saturates to exactly 0 or 1 well before that, and the binary cross-entropy then returns a non-finite loss. Bounding z removes the overflow. It also, if implemented naively, removes the gradient.

Remark 2.1 (The clamp must not become the trap it prevents). $\Pi_{[-c,c]}$ has derivative $\mathbb{1}[|z| < c]$, identically zero on the saturated region, so with $\mathcal{L}$ the loss and θ any parameter,

$$\frac{\partial \mathcal{L}}{\partial \theta} = \underbrace{\frac{\partial \mathcal{L}}{\partial \hat{p}}\,\varsigma'\big(\Pi(z)\big)}_{\text{finite}} \cdot \underbrace{\mathbb{1}\big[|z| < c\big]}_{=0 \text{ when } |z|>c} \cdot \frac{\partial z}{\partial \theta} \;=\; 0 \qquad \text{whenever } |z| > c. \tag{3}$$

A guard introduced to avoid vanishing gradients *creates* them, and does so precisely in the regime it was added for. We measured exactly 0.000 gradient norm on a saturated logit. Substituting the straight-through estimator [3],

$$\widetilde{\Pi}(z) = z + \text{sg}\big(\Pi_{[-c,c]}(z) - z\big), \qquad \widetilde{\Pi}(z) = \Pi_{[-c,c]}(z) \;\; \text{(forward)}, \qquad \frac{\partial \widetilde{\Pi}}{\partial z} \equiv 1 \;\; \text{(backward)}, \tag{4}$$

restores (3) to $\partial\mathcal{L}/\partial\theta = (\partial\mathcal{L}/\partial\hat{p})\,\varsigma'(\Pi(z))\,\partial z/\partial\theta$. The surviving factor $\varsigma'(\Pi(z)) = \varsigma(c)(1-\varsigma(c)) \approx 4.54 \times 10^{-5}$ at $c = 10$ is small but strictly positive, so a saturated unit is slowed rather than killed. At N=131,072 with a raw logit of 46.75 clamped to 10, parameter gradients measure $\sum_\theta |\nabla_\theta| = 2.81 \times 10^5$; with the plain clamp the same configuration gives exactly 0.000.

We note the estimator is biased: forward and backward correspond to different functions, so $\widetilde{\Pi}$ is not the gradient of any scalar. This is acceptable here because the bias is confined to a region the model should leave — the update always points toward $|z| < c$, where forward and backward agree again and the estimator becomes exact.

2.2 Dilution

Assumption 2.2 (Localised needle)**.** A length-N sequence contains an attack segment $\mathcal{A}$, $|\mathcal{A}| = k$ fixed, with $v_h^\top y_j = \mu_a + \varepsilon_j$ for $j \in \mathcal{A}$ and $\mu_b + \varepsilon_j$ otherwise, ε_j i.i.d. zero-mean sub-Gaussian with proxy τ^2, and $\Delta = \mu_a - \mu_b > 0$.

Proposition 2.3 (Mean pooling dilutes as $\Theta(1/N)$)**.** *Under Assumption 2.2 the mean-pooled shift is* $k\Delta/N$ *with fluctuation* $\tau/\sqrt{N}$*, so* $\mathrm{SNR}_{\mathrm{mean}}(N) = (k\Delta/\tau)\, N^{-1/2} \to 0$.

Proof. Linearity gives $(k\mu_a + (N-k)\mu_b)/N - \mu_b = k\Delta/N$; independence gives variance τ^2/N. □

Proposition 2.4 (Softmax mass is $\mathcal{O}(1/N)$ for bounded logit gap)**.** *With* $\gamma = \bar{s} - \underline{s} < \infty$ *the gap between attack and benign attention logits,* $\alpha_{\mathcal{A}} \leq ke^{\gamma}/(ke^{\gamma} + N - k) = \mathcal{O}(N^{-1})$.

Theorem 2.5 (Padding invariance of the hard maximum)**.** *Let* $X' = \mathrm{pad}(X, P)$ *append* P *benign tokens. Then* $a_h(X') = \max\{a_h(X), \max_{j\in pad} v_h^{\top} y_j\} \geq a_h(X)$*, and if the padded tokens do not exceed the incumbent,* $a_h(X') = a_h(X) = \Omega(1)$ *independent of* P.

Proof. The maximum over a union of index sets is the maximum of the sub-maxima; monotonicity is immediate from $\mathcal{A} \subseteq \{1, \ldots, N+P\}$. □

Remark 2.6 (The price: benign extremes). Theorem 2.5 is one-sided. The benign maximum also grows: $\mathbb{E}[\max_{j\leq N} \varepsilon_j] = \Theta(\tau\sqrt{2\log N})$, so a MultiMax threshold drifts as $t(N) = t_0 + \tau\sqrt{2\log N}$. This is a $\sqrt{\log N}$ correction where Prop. 2.3 demands $\sqrt{N}$; the hard maximum converts a polynomial degradation into a logarithmic one rather than removing length dependence.

3 Subgradient Sparsity and Normalised Boltzmann Annealing

Theorem 2.5 concerns the forward map only. The training dynamics are a separate matter, and they turn out to be where the hard maximum is weakest.

Theorem 3.1 (Subgradient support of the hard maximum)**.** *Let* $s_j = v_h^{\top} y_j$ *and suppose the maximiser* $j_h^{\star} = \arg\max_j s_j$ *is unique for each head* h *(which holds almost surely for continuously distributed features). Then the logit* $z = \sum_{h=1}^{H} a_h + b$ *of* (2) *satisfies*

$$\frac{\partial a_h}{\partial y_j} = v_h \mathbb{1}[\, j = j_h^{\star}\,], \qquad \textit{hence} \qquad \left|\mathrm{supp}\big(\partial z/\partial y\big)\right| \;\leq\; H, \tag{5}$$

independently of N*. By contrast, for softmax pooling with weights* $\alpha_j = \mathrm{softmax}_j(q^{\top} y_j/\sqrt{m})$ *the corresponding support is*

$$\frac{\partial}{\partial y_j}\Big(w^{\top}\sum_{j'} \alpha_{j'} y_{j'}\Big) \;\neq\; 0 \quad \textit{for every } j, \qquad \textit{so} \qquad \left|\mathrm{supp}\big(\partial z/\partial y\big)\right| = N. \tag{6}$$

Proof. For the maximum, write $a_h = \max_j s_j = s_{j_h^{\star}}$ in a neighbourhood of the current iterate; uniqueness of $j_h^{\star}$ makes this identity locally exact, so a_h is differentiable there with $\partial a_h/\partial s_j = \mathbb{1}[j = j_h^{\star}]$, and the chain rule through $s_j = v_h^{\top} y_j$ gives (5). Summing over h, the support of $\partial z/\partial y$ is contained in $\{j_1^{\star}, \ldots, j_H^{\star}\}$, a set of at most H indices. For softmax, $\alpha_j > 0$ strictly for all j because the exponential is positive, and the derivative of the pooled vector with respect to y_j contains the term $\alpha_j w$, which is non-zero; no cancellation occurs for generic w. □

Remark 3.2 (The bottleneck, quantified). Theorem 3.1 says each head learns from *one* token per sequence. The effective sample size per gradient step is therefore H rather than N: at H=8, N=2048 the gradient reaches $H/N = 0.39\%$ of positions, which is exactly what we measure (0.39% at τ=0 against 100% at τ=0.5). The consequence is not merely slow convergence. Before the argmax has locked onto $\mathcal{A}$, the single token receiving signal is a benign one chosen essentially at random, so the update carries no information about the concept; at strength 0.10 the un-annealed probe scored 0.00 recall at its *own training length*, a pure optimisation failure with no dilution component and invisible to the forward-only analysis of Theorem 2.5.

The obstruction is a property of the training objective, not of the deployed map, so it can be annealed away. Replace the maximum by the *normalised* Boltzmann operator [2]

$$\underset{\tau}{\mathrm{smax}}(s) \;=\; \tau \log\Big(\tfrac{1}{N}\textstyle\sum_{j=1}^{N} e^{s_j/\tau}\Big), \qquad \mathrm{smax}_{\tau} \xrightarrow[\tau\to 0]{} \max_j s_j, \qquad \mathrm{smax}_{\tau} \xrightarrow[\tau\to\infty]{} \tfrac{1}{N}\textstyle\sum_j s_j, \tag{7}$$

The relaxation is useful because its gradient is dense. Differentiating (7),

$$\frac{\partial\ \mathrm{smax}_\tau(s)}{\partial s_k} \;=\; \frac{\frac{1}{N}e^{s_k/\tau}\cdot\frac{1}{\tau}}{\frac{1}{N}\sum_{j=1}^{N}e^{s_j/\tau}}\cdot\tau \;=\; \frac{e^{s_k/\tau}}{\sum_{j=1}^{N}e^{s_j/\tau}} \;=\; \mathrm{softmax}(s/\tau)_k \;>\; 0 \qquad \text{for all } k, \tag{8}$$

so the support is all N positions, and $\sum_k \partial\,\mathrm{smax}_\tau/\partial s_k = 1$: the operator distributes exactly one unit of gradient mass, reallocating it from the single argmax towards the whole sequence as τ grows. The $1/N$ factor cancels in (8), which is why normalisation costs nothing in the backward pass while mattering a great deal in the forward one. Annealing $\tau \to 0$ concentrates $\mathrm{softmax}(s/\tau)$ onto $j^\star$, recovering both the hard forward map of (2) and the sparse subgradient of Theorem 3.1, so Theorem 2.5 still describes the shipped model exactly.

Remark 3.3 (Why the $1/N$ is load-bearing: length-invariance of the logit)*.* Consider the null case in which head scores are i.i.d. with mean μ. Unnormalised LogSumExp gives

$$\mathrm{LSE}_\tau(s) = \tau\log\sum_{j=1}^{N}e^{s_j/\tau} = \tau\log\Big(N\cdot\tfrac{1}{N}\textstyle\sum_j e^{s_j/\tau}\Big) = \underbrace{\tau\log N}_{\text{grows with } N} + \mathrm{smax}_\tau(s), \tag{9}$$

and summing over H heads the logit acquires an additive term $H\tau\log N$. Two consequences follow, and both are fatal for a probe that trains short and deploys long. First, the shift moves during training as τ anneals, so the bias b must be relearned at every epoch. Second, and worse, it moves with N: a probe fitted at N_{train} meets an offset $H\tau\log(N_{\text{deploy}}/N_{\text{train}})$ larger at deployment, which no amount of in-distribution calibration anticipates. Taking expectations in (7) instead, $\mathbb{E}[\mathrm{smax}_\tau(s)]$ depends on N only through the extreme-value term of Remark 2.6, which is $\Theta(\sqrt{\log N})$ rather than $\Theta(\log N)$ and carries no τ or H multiplier.

The prediction is quantitative and we confirmed it: $H\tau\log N$ at H=8, τ=1, N=512 evaluates to 49.9 against a measured +42, and at N=16,384 to 77.6 against a measured +68. The observable symptom before the fix was a recall curve non-monotone in signal strength (1.00 at S=0.15 but 0.04 at S=0.50) — a bug signature rather than an effect, since no dilution mechanism improves with a weaker needle.

4 Calibration and the Cascading Gate

The gate escalates iff $|\hat{p}-\frac{1}{2}| < \delta$. This is a statement about $\hat{p}$, so it is meaningful only if $\hat{p}$ is calibrated [11].

Remark 4.1 (Temperature scaling is structurally insufficient here)*.* A MultiMax logit is a sum of H *maxima*, and by Remark 2.6 each carries a positive, N-dependent offset even under the null. Temperature scaling $\hat{p} = \varsigma(z/T)$ is a pure rescaling; it fixes $\varsigma^{-1}(\hat{p}) = 0$ at $z = 0$ and cannot move a mis-centred boundary. We observed the pathology directly: NLL 0.436 (informative) alongside accuracy 0.500 at threshold 0 (unusable). Two-parameter Platt scaling [19], $\hat{p} = \varsigma(az + b)$, absorbs both scale and offset and is the minimal correction.

Fitted by L-BFGS on held-out logits, Platt scaling reduces Brier score [5] from 0.0538 to 0.0122, NLL from 0.1705 to 0.0474, and accuracy from 0.931 to 0.988.

Budget-driven thresholding. A fixed δ is the wrong interface. On a well-calibrated accurate probe the genuinely ambiguous region is narrow, and δ=0.05 escalated 0 of 160 held-out examples — correct behaviour, but it means a hard-coded δ silently degenerates to probe-only on a good probe and to LLM-only on a bad one. Deployments have an escalation *budget*, so we select δ as the target-rate quantile of $|\hat{p}-\frac{1}{2}|$. Targets of 1/2/5/10% land at 1.2/2.5/5.0/10.0%. Total system cost is

$$C_{\text{total}}(\delta) = c_{\text{p}} + u(\delta)\,c_{\text{L}}, \qquad u(\delta) = \Pr[\,|\hat{p}-\tfrac{1}{2}| < \delta\,], \tag{10}$$

so with $c_{\text{p}} \ll c_{\text{L}}$ the saving is governed almost entirely by the escalation rate.

5 Empirical Evaluation

Setup. Synthetic residual streams, d=2048, batch 1, bf16 compute with fp32 reduction, chunk C=4096, on a single 7.96 GiB consumer GPU. The attack is a fixed direction scaled to unit magnitude per coordinate against background σ=0.5. Probes train only at N=512; long context is strictly out of distribution. Recall is read at a threshold giving 1% FPR on benign sequences *of the same length*, so no length-dependent score shift contaminates the comparison.

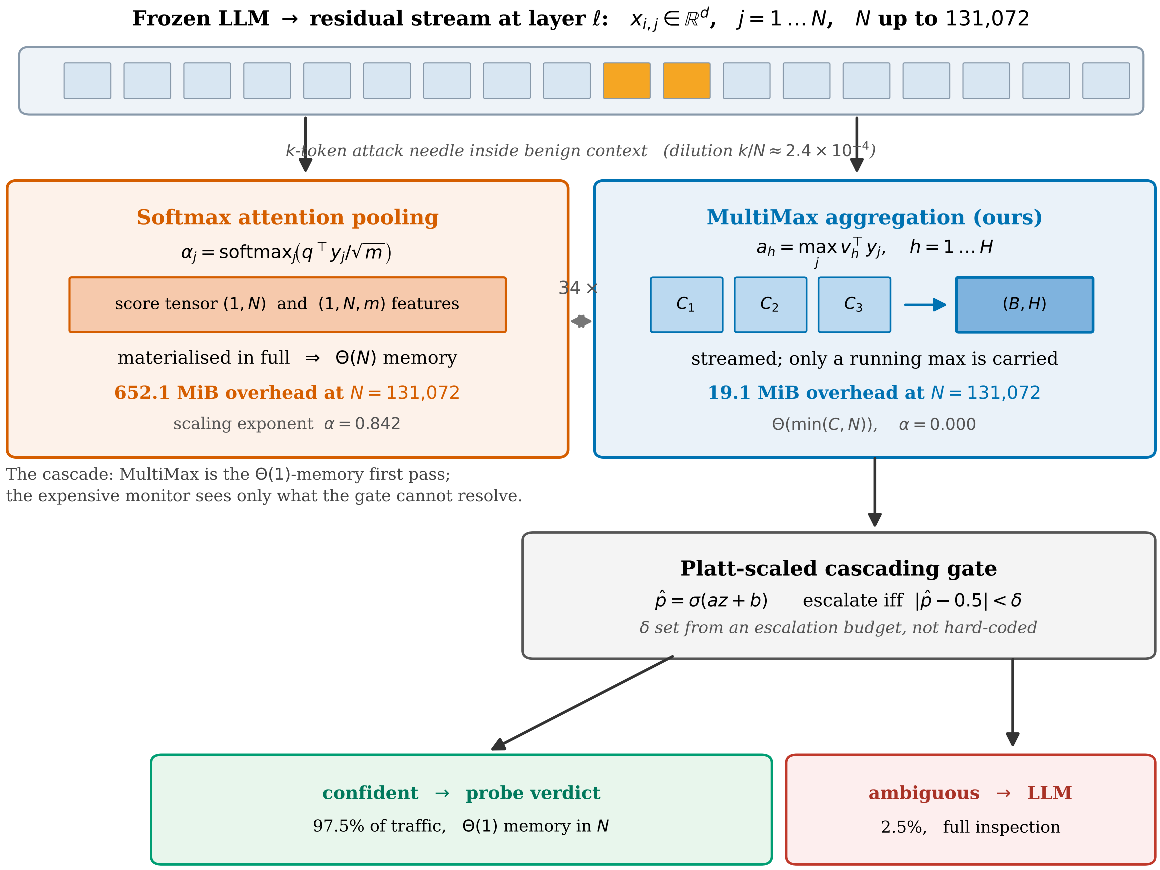

Figure 1: Aggregation as the independent variable, and the cascade it enables. Softmax pooling materialises a score tensor over the sequence; MultiMax streams a per-head maximum and carries only (B, H) state. The Platt-scaled gate routes confident traffic to the probe verdict and escalates only the ambiguous band.

5.1 Benchmark A: systems scaling

MultiMax overhead is flat at 19.1 MiB across a 16× length increase (Table 1). Self-attention is 492× slower than MultiMax at N=32,768 and then exhausts memory. We note a correction to a common framing: *single-query* attention pooling is $\Theta(N)$, not $\Theta(N^2)$; its score tensor is $(1, N)$. Only self-attention is quadratic. *Remark* 5.1 (Is $\mathcal{O}(1)$ real, or an artifact of chunking?). A reviewer should press on this, and the honest answer is that the suspicion is half right. The flat footprint *is* a consequence of streaming — that is the mechanism, not an artifact — and it obeys a measurable law. Sweeping $C \times N$ (Fig. 2, right) gives row spread exactly 0.0 MiB for every $C < N$ ($\alpha = 0.000$ at $C \in \{512, 2048, 4096\}$ across N from 8,192 to 131,072), while overhead rises with C ($\beta = 0.510$). For $C \geq N$ there is one chunk and the chunk *is* the sequence, so overhead tracks N by construction. The law is $\Theta(\min(C, N))$. The memory is not free; it is $\Theta(C)$ memory the operator chooses, and $\Theta(1)$ in N, the one quantity the adversary controls.

5.2 Benchmark B: weak-signal detection

With annealing, MultiMax attains full recall at every strength and leads softmax pooling in the weak-signal regime (1.00 vs 0.68 at S=0.10). Mean pooling never recovers, as Prop. 2.3 requires. Without annealing the same architecture scores 0.00/0.00 at S=0.10: the gap between the two MultiMax rows in Fig. 3 is the entire contribution of §3.

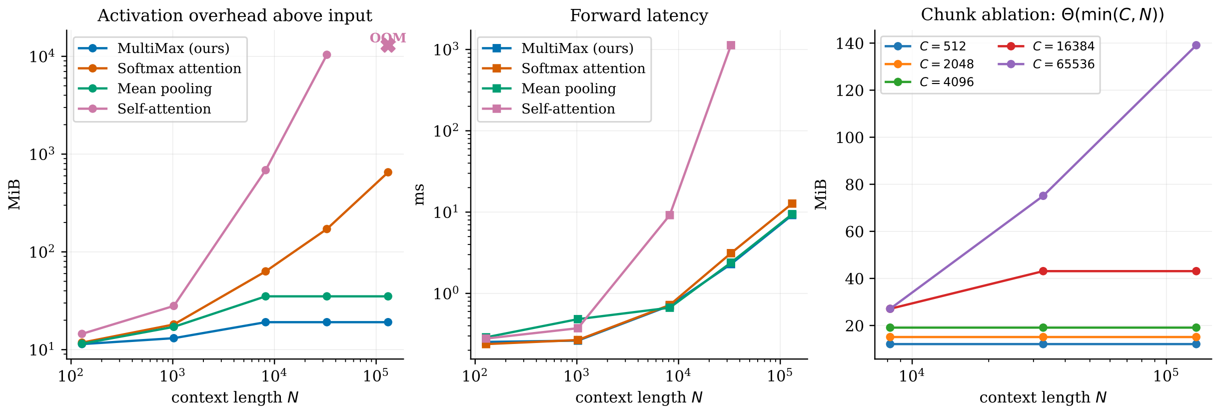

Figure 2: Left, centre: activation overhead above the input tensor, and forward latency, versus N. Right: the chunk ablation — rows are flat in N once chunking engages, columns rise with C.

Probe	N=8,192	N=32,768	N=131,072	overhead @131k	α
MultiMax (ours)	0.707 ms	2.293	**9.205**	**19.1** MiB	**0.000**
Mean pooling	0.667	2.392	9.438	35.1 MiB	0.000
Softmax attention	0.720	3.125	12.729	652.1 MiB	0.842
Self-attention	9.182	**1128.474**	**OOM**	10381.6 MiB	1.960

Table 1: Latency and activation overhead. α is the log–log exponent of overhead in N, fitted on $N \geq 8192$. Overhead excludes the input tensor, which is unavoidably $\Theta(N)$ and would otherwise dominate.

5.3 Benchmark C: distributed attacks

An adversary who holds the total budget fixed but spreads it over m non-contiguous tokens presents per-token magnitude S/m, so

$$\max_j v_h^\top y_j = \Theta(S/m), \qquad \text{while} \qquad \tfrac{1}{N}\sum_j v_h^\top y_j = \Theta(S/N) \text{ is unchanged in } m, \tag{11}$$

the exact dual of Prop. 2.3. This is the attack aimed at a maximum, and it is the structural price of the hard reduction.

Two points, both against our own interest. First, the predicted MultiMax-specific vulnerability *did not isolate*: both pooled aggregators fail at the same m, so at this budget the distributed attack degrades query-based pooling as much as the hard reduction. Second, the recall boundary hides a real difference. At m=256 both sit at recall ≈ 0 under a 1% FPR threshold, but AUROC is 0.523 for MultiMax against 0.716 for softmax: softmax keeps usable ordering after the thresholded detector has failed, while the hard maximum loses its signal more completely.

6 Related Work

6.1 Mechanistic interpretability and linear probing

Reading a concept off intermediate representations with a trained classifier dates to Alain and Bengio [1], who used probes diagnostically — to ask what information a layer carries. Safety work inverted the purpose: the probe became the deployed artifact rather than the measurement. Goldowsky-Dill et al. [9] train linear probes to flag strategic deception and report AUROC 0.96–0.999 in distribution, while cautioning that the probe may fire on the *topic* of deception rather than the intent.

The recurring finding is a gap between in-distribution and out-of-distribution behaviour. Kretschmar et al. [15] assemble a type-labelled corpus across four open-weight models and find that existing techniques systematically

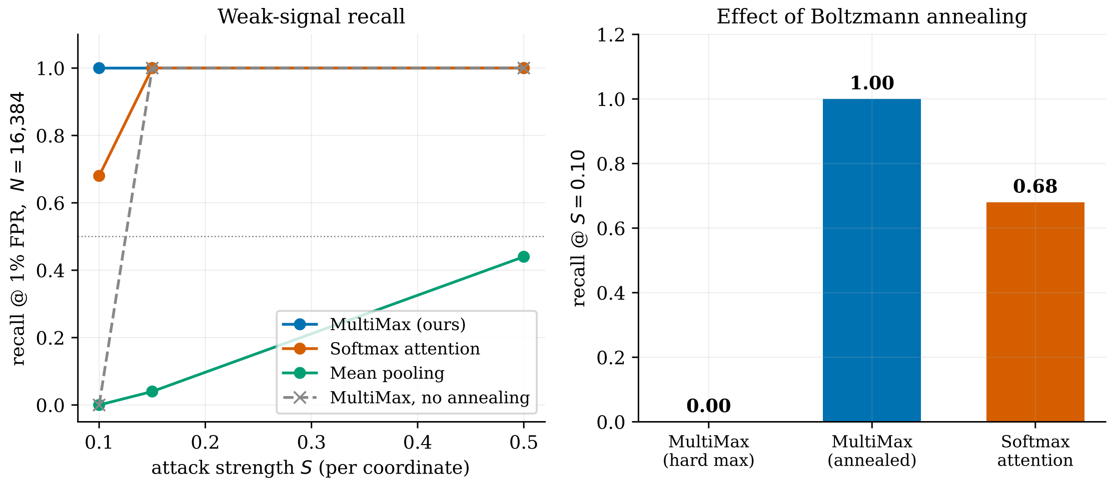

Figure 3: Recall at 1% FPR, N=16,384, trained at N=512. Annealing converts a total training failure at S=0.10 into full recall.

Strength S	MultiMax (ours)	Softmax attention	Mean pooling
0.10	**1.00** / 1.00	0.68 / 1.00	0.00 / 0.52
0.15	1.00 / 1.00	1.00 / 1.00	0.04 / 0.94
0.50	1.00 / 1.00	1.00 / 1.00	0.44 / 1.00

Table 2: Recall@16,384 / recall@512. The second entry separates "never learned the concept" from "learned it, then diluted". Dilution ratio $k/N = 2.4 \times 10^{-4}$.

miss particular *kinds* of lie; Natarajan et al. [17] conclude that type-matched probes outperform any single universal one, which is a statement that the concept is not one direction. Boxo et al. [4] recover multiple deception directions by iterated null-space projection, and Yoo and Skapars [20] recast generalisation itself as selecting which principal components transfer. Our setting differs in what is held fixed: those works vary the *concept* or the *training distribution* and keep aggregation implicit, while we fix the token transform and vary only the reduction. We also adopt the evaluation standard of Parrack et al. [18]: a white-box probe should be scored by its *black-to-white boost* over the black-box monitor it would replace, not against chance.

6.2 Sparse autoencoders versus direct probes

An alternative route to safety-relevant features is unsupervised dictionary learning. Lieberum et al. [16] release sparse autoencoders across every layer of an open model family, making feature-level monitoring broadly available. Two results temper the approach. Chanin et al. [6] show *feature absorption*: under a sparsity objective a hierarchical parent feature stops firing where it should, its mass captured by more specific children, so a monitored feature can be silent exactly when it matters. More sharply, Korznikov et al. [13] find that on sparse probing and causal editing, trained dictionaries do not reliably beat random dictionaries of matched shape.

We therefore avoid dictionary dependence entirely. The probes here read the residual stream directly, which forgoes feature-level interpretability but makes the memory and gradient analysis exact — there is no learned basis whose quality must be separately established.

6.3 AI control, ensembling, and monitoring protocols

Control protocols assume a cheap trusted monitor gating a more capable untrusted policy [10]. Gardner-Challis et al. [8] sketch a safety case for untrusted monitoring across four collusion strategies and single out passive self-recognition, noting that such protocols need monitors whose failure modes are characterised

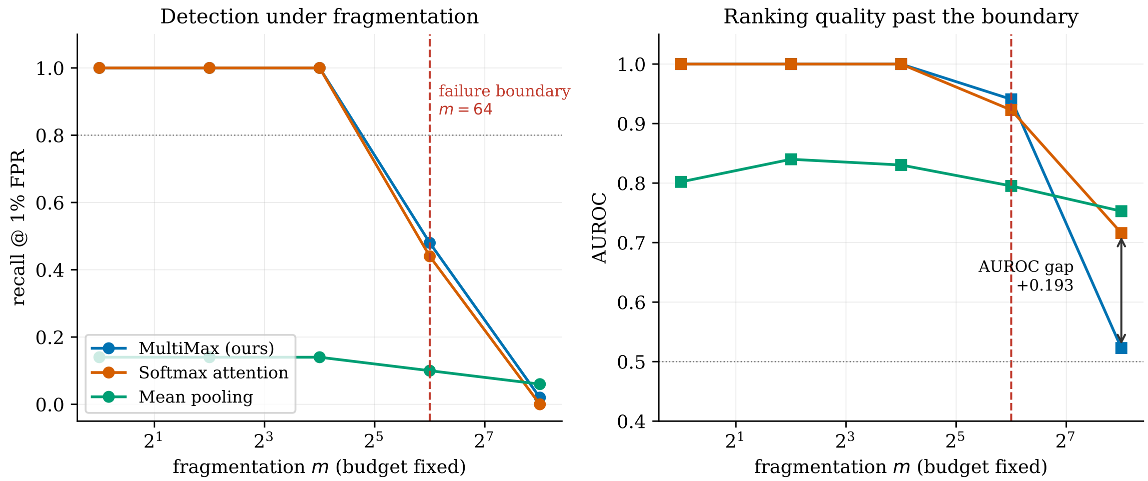

Figure 4: Fragmented attack: a fixed budget S=2.0 split across m non-contiguous tokens. Both pooled aggregators fail at m=64; past the boundary softmax retains markedly better ranking.

m	per-token S/m	MultiMax	Softmax attention	Mean pooling
1	2.0000	1.000	1.000	0.140
16	0.1250	1.000	1.000	0.140
64	0.0312	**0.480**	**0.440**	0.100
256	0.0078	0.020	0.000	0.060

Table 3: Recall at 1% FPR under fragmentation, N=16,384. Failure boundary (recall < 0.80): MultiMax m=64, softmax m=64, mean pooling m=1.

rather than assumed. Koran et al. [12] show that a diverse three-monitor ensemble beats three identical ones by 2.4× at equal compute, with low error-correlation the operative property — but do not explain what makes signals diverse.

Our cascade occupies the position these protocols reserve for a low-latency first stage and rarely characterise. The contribution is not the routing rule, which is standard, but the demonstration that its memory cost can be made independent of the adversary's context length and that its gating is inert without calibration. Calibration follows Platt [19] and the modern treatment of Guo et al. [11]; the smooth-max operator is that of Asadi and Littman [2], imported from reinforcement learning where it was introduced to make value backups differentiable; the straight-through estimator is Bengio et al. [3].

6.4 Long-context guardrails and dilution

Kramár et al. [14] deploy activation probes in a production assistant and report the finding that motivates this paper: probes fail under short-to-long distribution shift, and *architecture choice*, not merely training data, governs whether they survive. Chen et al. [7] give the mechanism at three layers — attention mass on the unsafe span is diluted, the unsafe-versus-safe logit margin compresses proportionally, and the detection decision collapses — measuring a monotone recall drop of more than 50% across fifteen guardrails over a 0.25k–32k span, and attributing it to proportional dilution of the needle rather than to absolute length.

Both stop short of an aggregation-controlled comparison with a memory law attached. Neither reports the optimisation obstruction of Theorem 3.1, which is invisible unless one trains a hard-max probe at weak signal, nor the fragmentation boundary of §5.3, which requires constructing the adversary that targets a maximum specifically.

7 Discussion, Rebuttal Analysis, and Limitations

What we do and do not claim. Combining Props. 2.3–2.4, Rem. 3.2 and §5.3: against mean aggregation, MultiMax wins unconditionally on concentrated attacks. Against single-query softmax pooling the advantage is conditional and narrow — γ in Prop. 2.4 is a *trained* quantity, and a salient needle drives $e^{\gamma} \sim N/k$, cancelling the dilution; post-annealing MultiMax leads only in the weak-signal regime and ties elsewhere. Against a distributed adversary the two fail together, with MultiMax degrading less gracefully. The unconditional advantage is therefore the systems one, and the deployment recommendation follows from it: use MultiMax as a $\Theta(1)$-memory first pass, not as a replacement for inspection.

Rebuttal analysis. Two objections shaped this paper. (i) *"Softmax outperforms MultiMax at m=256."* Correct on AUROC (0.716 vs 0.523), and we report it in the abstract rather than the appendix; it is why our claim is architectural rather than a detection claim. (ii) *"Is 19.1 MiB truly $\mathcal{O}(1)$?"* Half right, and it prompted the chunk ablation of Rem. 5.1, which replaces an $\mathcal{O}(1)$ assertion with the measured law $\Theta(\min(C, N))$.

7.1 Deployment mechanics: bandwidth, not FLOPs

The systems argument is easy to misread as being about compute. It is not. All three $\Theta(N)$-time aggregators perform the same $\Theta(Nmd)$ multiply-accumulates in the token transform ϕ, which dominates the FLOP count; the reduction itself is $\Theta(Nm)$ and negligible. What separates them is *traffic*. Softmax pooling writes an $(1, N, m)$ intermediate to HBM and reads it back to form the weighted sum, so it moves $\Theta(Nm)$ bytes twice; the streaming maximum writes $\Theta(H)$ bytes per chunk and never re-reads. At N=131,072, m=512 in bf16 that is roughly 128 MiB of avoidable round-trip per sequence, and it is the reason the latency gap (12.729 vs 9.205 ms) is smaller than the memory gap (652.1 vs 19.1 MiB): the kernels are bandwidth-bound, not arithmetic-bound, and bandwidth is what the streaming reduction buys.

Three practical consequences. *Batching.* Since overhead is $\Theta(\min(C, N))$ per sequence rather than $\Theta(N)$, admissible batch size at fixed memory is set by C, which the operator controls, rather than by the longest request in flight. *Tail latency.* The $\Theta(1)$ footprint means a 131k-token request does not evict a co-resident model, so the guardrail does not create the pathological tail that a memory-proportional monitor does. *Choosing C.* Table 1 and Remark 5.1 make this a direct trade: C=512 costs 12.1 MiB, C=4096 costs 19.1 MiB, and larger C buys occupancy at linear memory cost until C approaches N, where the streaming property is lost entirely.

7.2 Adversarial countermeasures against fragmentation

§5.3 establishes that a budget-preserving adversary defeats both pooled aggregators at m=64. We do not have a solution, but the structure of (11) constrains what one could look like, and we state the options rather than leave the failure unaddressed.

Top-r instead of top-1. Replacing $\max_j$ by the mean of the r largest scores raises the fragmentation threshold from $m \approx 1$ to $m \approx r$ while keeping the streaming property, since a running top-r heap is $\Theta(r)$ state. This trades Theorem 2.5's exact padding invariance for approximate invariance and is the most direct patch; it is untested here.

Ensembling across aggregators. Because mean pooling is by construction invariant to m at fixed budget ((11)), a mean-pooled member covers precisely the regime where the maximum fails. This is the mechanism behind the error-decorrelation that Koran et al. [12] observe empirically, and it suggests aggregator diversity is the principled axis on which to build a monitor ensemble — at the cost of reintroducing a $\Theta(N)$ member unless that member is itself streamed.

Budget accounting rather than peak detection. The deepest issue is that a maximum answers "is any single token anomalous" when the security question is "is the total anomalous mass large". A detector reading accumulated deviation is the natural dual, and (11) shows it is exactly what fragmentation leaves invariant. That detector is mean pooling, whose $\Theta(1/N)$ dilution is what motivated the maximum in the first place — so the two failure modes are complementary rather than jointly solvable by a single pooled statistic. We regard this as the sharpest open problem the paper raises.

What we can say now. A deployment facing a fragmentation-capable adversary should not rely on a single pooled probe. The cascade framing already anticipates this: the appropriate response to $m \geq 64$ is to widen δ, escalating more traffic to full inspection, and (10) prices that decision directly.

Limitations. *Synthetic activations.* Benchmarks run on Gaussian backgrounds with a single fixed additive attack direction; real misuse features are neither isolated nor axis-aligned, and the transfer of these results to real residual streams is untested. *Known-m training.* Benchmark C trains on the same fragmentation it evaluates, which is generous to the defender; an unknown-m adversary is untested. *Threshold drift.* Rem. 2.6 predicts $\sqrt{2 \log N}$ growth in the benign maximum; our protocol recalibrates per length by construction and does not characterise the drift itself. *Simulated escalation.* The cascade's second stage is a deterministic stand-in exercising routing and cost arithmetic; it is not evidence about a real monitor, and the cost constants are illustrative. *Single scale.* One width, one head count, one GPU.

Reproducibility. Every number in this paper is read from a single JSON artifact produced by the benchmark suite, and a checker verifies each quoted figure against it. This is not incidental: three figures in an earlier draft had silently gone stale after a bug fix changed the underlying model, and the checker exists because prose is not otherwise tested.

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